

Assignment # 03

6.1 (LAPLACE TRANSFORMS)

Q1) $3t + 12$

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$$

$$\mathcal{L}(t) = 1/s^2$$

$$\mathcal{L}(1) = 1/s$$

Using linear property:

$$\mathcal{L}\{af(t) + bg(t)\} = a\mathcal{L}\{f(t)\} + b\mathcal{L}\{g(t)\}$$

$$\mathcal{L}\{3t\} = 3\mathcal{L}\{t\} = \frac{3}{s^2}$$

$$\mathcal{L}\{12\} = 12\mathcal{L}\{1\} = \frac{12}{s}$$

$$\mathcal{L}\{3t + 12\} = \frac{3}{s^2} + \frac{12}{s}$$

$$Q_3) \cos \pi t$$

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$$

$$\mathcal{L}\{\cos(\pi t)\} = \int_0^{\infty} e^{-st} \cos(\pi t) dt$$

To solve this integral, we use the standard result for the Laplace transform of $\cos(\omega t)$:

$$\mathcal{L}\{\cos(\omega t)\} = \frac{s}{s^2 + \omega^2}$$

Substituting ω with π , we get

$$\mathcal{L}\{\cos(\pi t)\} = \frac{s}{s^2 + \pi^2}$$

INVERSE LAPLACE TRANSFORMS

6.1

$$25) \frac{0.2s + 1.8}{s^2 + 3.24}$$

$$\frac{0.2s + 1.8}{s^2 + (\sqrt{3.24})^2} = \frac{0.2s + 1.8}{s^2 + (1.8)^2}$$

$$= \frac{0.2s}{s^2 + (1.8)^2} + \frac{1.8}{s^2 + 1.8^2}$$

$$\because \frac{s}{s^2 + \omega^2} = \cos \omega t$$

$$\because \frac{\omega}{s^2 + \omega^2} = \sin(\omega t)$$

In our case, $\omega = 1.8$

$$\mathcal{L}^{-1} \left\{ \frac{0.2s}{s^2 + 1.8^2} \right\} = 0.2 \cos(1.8t)$$

$$\mathcal{L}^{-1} \left\{ \frac{1.8}{s^2 + 1.8^2} \right\} = \sin(1.8t)$$

Combine

$$\mathcal{L}^{-1} \left\{ \frac{0.2s + 1.8}{s^2 + (1.8)^2} \right\} = 0.2 \cos(1.8t) + \sin(1.8t)$$

6.1

$$Q_{27}) \frac{s}{L^2 s^2 + n^2 \pi^2}$$

$$\frac{s}{L^2 s^2 + n^2 \pi^2} = \frac{s}{L^2 \left(s^2 + \frac{n^2 \pi^2}{L^2} \right)}$$

$$= \frac{s}{L^2 \left(s^2 + \left(\frac{n\pi}{L} \right)^2 \right)}$$

$$\therefore \mathcal{L}^{-1} \left[\frac{s}{s^2 + \omega^2} \right] = \cos(\omega t)$$

Here, $\omega = \frac{n\pi}{L}$, Therefore, we have.

$$\mathcal{L}^{-1} \left\{ \frac{s}{L^2 \left(s^2 + \left(\frac{n\pi}{L} \right)^2 \right)} \right\} = \frac{1}{L^2} \cos \left(\frac{n\pi}{L} t \right)$$

$$\boxed{\frac{1}{L^2} \mathcal{L}^{-1} \left\{ \frac{s}{s^2 + \left(\frac{n\pi}{L} \right)^2} \right\} = \frac{1}{L^2} \cos \left(\frac{n\pi}{L} t \right)}$$

APPLICATION OF S-SHIFTING

$$33) t^2 e^{-3t}$$

$$\mathcal{L}\{e^{at} f(t)\} = f(s-a)$$

$$\text{where } F(s) = \mathcal{L}\{f(t)\}$$

$$\mathcal{L}\{t^2\} = \int_0^{\infty} e^{-st} t^2 dt$$

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$$

For $n=2$:

$$\mathcal{L}\{t^2\} = \frac{2!}{s^{2+1}} = \frac{2}{s^3}$$

Now, using the s-shifting property for $t^2 e^{-3t}$:

$$\mathcal{L}\{t^2 e^{-3t}\} = \mathcal{L}\{t^2\} \Big|_{s \rightarrow s+3}$$

Substitute s with $s+3$ in $2/s^3$:

$$\boxed{\mathcal{L}\{t^2 e^{-3t}\} = \frac{2}{(s+3)^3}}$$

31) $\pi / (s + \pi)^2$

This can be recognized in the form of $K / (s - a)^n$ which is a known Laplace transform for $t^{n-1} e^{at} / \Gamma(n)$. Specially for $n=2$ and $a = -\pi$, the inverse Laplace transform is related to the function $t e^{-\pi t}$.

The inverse Laplace transform of $\frac{1}{(s-a)^2}$ is $t e^{at}$:

$$\mathcal{L}\{e^{at} f(t)\} = F(s-a)$$

$$\mathcal{L}^{-1}\left\{\frac{1}{(s-0)^2}\right\} = t$$

$$\mathcal{L}^{-1}\left\{\frac{1}{(s+\pi)^2}\right\} = t e^{-\pi t}$$

$$\boxed{\mathcal{L}^{-1}\left\{\frac{\pi}{(s+\pi)^2}\right\} = \pi t e^{-\pi t}}$$

6.3

$$t-2 \quad (t > 2)$$

The function $f(t) = t-2$ ~~sat~~ starts at $t=2$ and continues linearly. for $t < 2$, the function is zero.

$u(t-2)$ is 0 for $t < 2$ and 1 for $t \geq 2$. Thus, we can represent $f(t)$ as:

$$f(t) = (t-2)u(t-2)$$

$$\begin{aligned} \mathcal{L}\{(t-2)u(t-2)\} &= e^{-2s} \mathcal{L}\{t\} \\ &= e^{-2s} \cdot \frac{1}{s^2} \end{aligned}$$

$$\boxed{\mathcal{L}\{(t-2)u(t-2)\} = \frac{e^{-2s}}{s^2}}$$

$$5) e^t \quad (0 < t < \pi/2)$$

We can represent this function using unit step function:

$$f(t) = e^t [u(t) - u(t - \pi/2)]$$

$$\mathcal{L}\{e^t [u(t) - u(t - \pi/2)]\} = \mathcal{L}\{e^t u(t)\} - \mathcal{L}\{e^t u(t - \pi/2)\}$$

$$\mathcal{L}\{e^t u(t)\} = \mathcal{L}\{e^t\} = \frac{1}{s-1}$$

$$\mathcal{L}\{e^t u(t - \pi/2)\} = e^{-\pi/2 s} \mathcal{L}\{e^t |_{t=t+\pi/2}\}$$

$$\text{Let } \tau = t - \frac{\pi}{2} :$$

$$\begin{aligned} e^{-\pi/2 s} \mathcal{L}\{e^{\tau + \pi/2}\} &= e^{-\pi/2 s} \mathcal{L}\{e^{\tau} e^{\pi/2}\} \\ &= e^{-\pi/2 s} e^{\pi/2} \mathcal{L}\{e^{\tau}\} \end{aligned}$$

$$\text{Since } \tau = t - \pi/2 :$$

$$\mathcal{L}\{e^{\tau}\} = \frac{1}{s-1}$$

Therefore,

$$\mathcal{L}\left\{e^t u\left(t - \frac{\pi}{2}\right)\right\} = e^{-\frac{\pi}{2} s} e^{\pi/2} \frac{1}{s-1} = \frac{e^{-\pi/2(s-1)}}{s-1}$$

Combine:

$$\mathcal{L}\{e^t[u(t) - u(t - \pi/2)]\} = \frac{1}{(s-1)} \left(1 - e^{-\frac{\pi}{2}s + \pi/2} \right)$$

1) $\cos 4t \quad (0 < t < \pi)$

Using unit step function:

$$f(t) = \cos(4t) [u(t) - u(t - \pi)]$$

$$\begin{aligned} \mathcal{L}\{\cos(4t) [u(t) - u(t - \pi)]\} &= \\ \mathcal{L}\{\cos(4t) u(t)\} - \mathcal{L}\{\cos(4t) u(t - \pi)\} \end{aligned}$$

$$\mathcal{L}\{\cos(4t) u(t)\} = \mathcal{L}\{\cos(4t)\} = \frac{s}{s^2 + 16}$$

$$\mathcal{L}\{\cos(4t) u(t - \pi)\} = e^{-\pi s} \mathcal{L}\{\cos(4(t + \pi))\}$$

Since, $\cos(4(t + \pi)) = \cos(4t + 4\pi) = \cos(4t)$:

$$\mathcal{L}\{\cos(4t) u(t - \pi)\} = e^{-\pi s} \frac{s}{s^2 + 16}$$

Combine:

$$\mathcal{L}\{\cos(4t) [u(t) - u(t - \pi)]\} = \frac{s}{s^2 + 16} - e^{-\pi s} \frac{s}{s^2 + 16}$$

11.9 (Pg: 513-514)

Example : 2

Find the fourier integral representation of the function

$$f(x) = \begin{cases} 1 & \text{if } |x| < 1 \\ 0 & \text{if } |x| > 1 \end{cases}$$

$$\begin{aligned} A(\omega) &= \frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \cos \omega v \, dv = \frac{1}{\pi} \int_{-1}^1 \cos \omega v \, dv = \left. \frac{\sin \omega v}{\pi \omega} \right|_{-1}^1 \\ &= \frac{2 \sin \omega}{\pi \omega} \end{aligned}$$

$$B(\omega) = \frac{1}{\pi} \int_{-1}^1 \sin \omega v \, dv = 0$$

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \frac{\cos \omega x \sin \omega}{\omega} \, d\omega$$

$$\int_0^{\infty} \frac{\cos \omega x \sin \omega}{\omega} \, d\omega = \begin{cases} \pi/2 & \text{if } 0 \leq x < 1, \\ \pi/4 & \text{if } x = 1, \\ 0 & \text{if } x > 1. \end{cases}$$

$$\int_0^{\infty} \frac{\sin \omega}{\omega} \, d\omega = \frac{\pi}{2}$$

$$\text{Si}(u) = \int_0^u \frac{\sin w}{w} dw$$

$$\frac{2}{\pi} \int_0^a \frac{\cos wx \sin w}{w} dw$$

$$\frac{2}{\pi} \int_0^a \frac{\cos wx \sin w}{w} dw = \frac{1}{\pi} \int_0^a \frac{\sin(w+wx)}{w} dw + \frac{1}{\pi} \int_0^a \frac{\sin(w-wx)}{w} dw$$

$$= \frac{1}{\pi} \int_0^{(x+1)a} \frac{\sin t}{t} dt - \frac{1}{\pi} \int_0^{(x-1)a} \frac{\sin t}{t} dt$$

$$\frac{1}{\pi} \text{Si}(a[x+1]) - \frac{1}{\pi} \text{Si}(a[x-1])$$